

SURENDRA GHENTYALA

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EDUCATION

Cornell University

Aug 2022-Present

Ph.D. in Computer Science

Advisor: Noah Stephens-Davidowitz

Research Focus: Complexity Theory and Cryptography

Overall GPA: 4.04

University of California, Santa Barbara

Sep 2019-Dec 2021

B.S. in Computer Science *cum laude*

Advisor: Tefvik Bultan

EXPERIENCE

Google Research

May 2026-

Hosted by Pritish Kamath, Pasin Manurangsi, and Ravi Kumar

Summer internship

Columbia University

May 2025-Jul 2025

Hosted by Mihalis Yannakakis, also collaborated with Toniann Pitassi

Summer collaboration

Simons Institute for the Theory of Computing at UC Berkeley

May 2024-Aug 2024

Hosted by Venkatesan Guruswami

Research in LLM watermarking using cryptography and error-correcting codes

PAPERS

The Structure of In-Place Space-Bounded Computation ¹

James Cook, Surendra Ghentiyala, Ian Mertz, Edward Pyne, Nathan Sheffield

In submission

Hierarchies within TFNP: building blocks and collapses

Surendra Ghentiyala, Zeyong Li

In submission

Range Avoidance, Arthur-Merlin, and TFNP

Surendra Ghentiyala, Zeyong Li, Noah Stephens-Davidowitz

Symposium on the Theory of Computing (STOC), 2026

Efficient derandomization of differentially private counting queries

Surendra Ghentiyala

Symposium on Simplicity in Algorithms (SOSA), 2026

¹Unless otherwise specified, all authors are listed in alphabetical order, as in the standard in theoretical computer science.

Downward self-reducibility in the total function polynomial hierarchy
Karthik Gajulapalli, Surendra Ghentiyala, Zeyong Li, Sidhant Saraogi
Symposium on Discrete Algorithms (SODA), 2026

New constructions of pseudorandom codes
Surendra Ghentiyala, Venkatesan Guruswami
International Conference on Randomization and Computation (RANDOM), 2025

The more the merrier! On total coding and lattice problems and the complexity of finding multicollisions
Huck Bennett, Surendra Ghentiyala, Noah Stephens-Davidowitz
16th Innovations in Theoretical Computer Science (ITCS), 2025

More basis reduction for linear codes: backward reduction, BKZ, slide reduction, and more
Surendra Ghentiyala, Noah Stephens-Davidowitz
International Conference on Approximation Algorithms for Combinatorial Optimization Problems (APPROX), 2024

Obtaining Information Leakage Bounds via Approximate Model Counting
Seemanta Saha*, **Surendra Ghentiyala***, Shihua Lu, Lucas Bang, Tevfik Bultan
Programming Language Design and Implementation (PLDI), 2023
(*equal contribution)

TALKS

Range Avoidance, Arthur-Merlin, and TFNP
Symposium on the Theory of Computing (STOC), 2026

Efficient derandomization of differentially private counting queries
Symposium on Simplicity in Algorithms (SOSA), 2026

Downward self-reducibility in the total function polynomial hierarchy
Symposium on Discrete Algorithms (SODA), 2026

The Structure of In-Place Space-Bounded Computation
Cornell Theory Seminar, 2025

New constructions of pseudorandom codes
International Conference on Randomization and Computation (RANDOM), 2025

The more the merrier! On total coding and lattice problems and the complexity of finding multicollisions
16th Innovations in Theoretical Computer Science (ITCS), 2025

More basis reduction for linear codes: backward reduction, BKZ, slide reduction, and more
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Obtaining Information Leakage Bounds via Approximate Model Counting
Programming Language Design and Implementation (PLDI), 2023

TEACHING EXPERIENCE

Graduate Teaching Assistant, Cornell University

Aug 2022-May 2023

- CS 4700/5700 Foundations of Artificial Intelligence
- CS 4120/5120 Introduction to Compilers

AWARDS & ACHIEVEMENTS

NSF Graduate Research Fellowships Program (GRFP) Honorable Mention

April 2024

PROFESSIONAL SERVICE

Subreviewer

2023-2024

- ASIACRYPT (2026)
- EUROCRYPT (2026)
- SODA (2026)
- TCC (2025)
- ICALP (2024)
- FOCS (2023)

Graduate Student Recruitment for Outreach Events

Aug 2023 - Present

- For Association of Computer Science (ACSU) research night
- For informational panel on applying to graduate school

Research Night

2023-Present

- Present cryptography research once a semester to undergraduate students interested in computer science research

SKILLS

Programming Languages

C++, Haskell, Java, Python